

Assignment: Strings

1. Write a C program to read a line of text from the user and reverse the words in place.
2. Find and print the longest word by considering the sentence entered by the user.
Example:
Input: "I love programming in C"
Output: programming
3. Write a C program to compare two strings without using any string library functions.
4. Write a C program to read a string from the user and sort its characters using bubble sort algorithm.
Example:
Input: "dbca"
Output: "abcd"
5. Write a C program to read a sentence and replace lowercase characters with uppercase and vice versa.
Example:
Input: "Hello World"
Output: "hELLO wORLD"
6. Write a C program to remove characters from a string except alphabets.
Example: Input: A3bc4de.com
Output: Abcdecom
7. Write a C program to ask the user to input a string and a substring. Count how many times the substring occurs (even overlapping).
Example:
String: "aaa"
Substring: "aa"
Output: 2
8. Write a C program to ask the user to enter a message and a shift value (key), and perform both:
Encryption & Decryption.
Example: Input: "abc", key: 2
Output: Encrypted: "cde"
Decrypted: "abc"

9. Write a C program to ask the user to enter an expression string. Check if an expression contains properly balanced and nested.

Example: Input: "{[()()]}"

Output: Balanced

10. Write a C program to Remove all adjacent duplicate characters recursively from a string.

Example: Input: "azxxzy"

Step 1: "azzy" → Step 2: "ay"

Output: "ay"

11. Write a C program to ask the user for a string. Return whether it can become a palindrome by removing at most one character.

Example:

Input: "abca"

Output: Yes (remove 'b' or 'c')

12. Write a C program to find and print the indexes of all vowels and how many times each vowel appears by entering a sentence from the user.

Example:

Input: "education"

Output:

Vowels: e(0), u(1), a(4), i(6), o(7)

Counts: a=1, e=1, i=1, o=1, u=1

13. A) Find the bug in below code and explain :

```
#include <stdio.h>

int main() {
    char str[5];
    strcpy(str, "Hello");
    printf("%s\n", str);
    return 0;
}
```

14. Will below code compile and run?

```
#include <stdio.h>

int main() {
    char *str = "Hello";
    str[0] = 'h';
    printf("%s\n", str);
}
```

```

        return 0;
    }

```

15. What is wrong here?

```

#include <stdio.h>
#include <string.h>
int main() {
    char str1[] = "abc";
    char str2[] = "abc";
    if (strcmp(str1, str2) = 0)
        printf("Equal\n");
    return 0;
}

```

16. Write a function to replace characters in a string based on a lookup table (mapping old characters to new).

17. Write a C program to encode a given string in Base64 format.

18. Write a program to implement strcmp. Use your string comparison function to verify the string is palindrome or not.

19. Write a user-defined function to implement strrchr, which returns the last occurrence of a character in a string. NOTE: use one loop only.

20. Create user-defined strstr, strchr, atoi(ascii to int), itoa(integer to ascii), strcat.

21. Write a program to search a sub-string(s2) in a larger string(s1) and perform the followings:

22. remove all occurrences of s2 in s1.

23. Hide all occurrences of s2 in s1.

24. Replace all occurrences of s2 with its reverse equivalent in s1.

25. Explain the use of const type-qualifier.

26. Justify the modification(writing) on const-qualified-variable.

27. Can a global const-qualified variable be modified through pointer? Why?

28. If above answer is no, then what should we do to allow the modification.

29. Count the number of words in each string.

30. Remove the n-th word in each string.

31. Count and print the words having consecutive vowels.

32. Supply two strings. Verify, if the 1st string, is a circular permutation of another string.